

Table of Contents

Executive Summary	3
1.1 Recommended Product	3
1.2 Adaptability	3
1.3 ESG	4
1.4 Integration into ERM Framework	4
Product Design	5
2.1 Benefit Structures	6
2.2 Coverage triggers	6
2.3 Exclusions	6
Summary of Pricing and Capital Modeling	7
Risk Assessment	8
Assumptions	9
Data and Data Limitations	10
Appendix	12

Executive Summary

Space. The great unknown. A frontier of infinite possibilities and areas to be explored. But with these infinite possibilities, there also comes infinite risks. To fully explore these regions of unlimited potential, the obstacles that can hinder our ability to conduct operations in these areas must be taken into consideration. This report provides various pricing models to the Cosmic Quarry Mining Company, which specializes in the mining of deep space metals, crucial for advanced manufacturing. These pricing models cover claims related to equipment failure, cargo loss, and business interruptions, over five-star systems. These systems are the Helionis Cluster, the Bayesia System, the Oryn Delta, the Zeta solar system, and the Epsilon solar system. With these pricing models considering the risks associated with each star system, the Cosmic Quarry Mining Company (CQMC) will be better informed on how to insure operations within the star systems they are currently present, as well as how to expand operations into new star systems.

1.1 Recommended Product

The product that we recommend being added to Galaxy General Insurance Company's portfolio is an insurance plan for the three hazard area coverages: Business Interruptions, Equipment Failure, and Cargo Loss. This plan considers the varying risks of each solar system that our company has insured previously, such as Helionis Cluster, Bayesia, and Oryn Delta, while also considering the risks of the solar systems that Cosmic Quarry Mining Corporation is extending to – Epsilon and Zeta. Our product also includes total expected costs and generates a 5% profit for Galaxy General.

1.2 Adaptability

Our recommended product can be tailored for the distinct resources and potential challenges of each solar system by offering different ranges for exclusions and coverage triggers. We have a default deductible for each single policy which can be adjusted to each solar systems' unique event threat levels. For example, Zeta's challenge of the astronomical bodies interfering with ships' navigation systems has a direct negative effect on cargo loss. We then might issue a higher deductible for our Cargo Loss coverage since Zeta is more prone to frequent and large claims. Since all the claim amounts, and the frequencies remain about the same no matter the solar system we saw it fit to represent all the solar systems in one pricing model. We tailored this model to include all the risks in one galaxy and since each galaxy has no significant predictors in estimating claim amount or claim frequency, due to their individual risks we compiled them all into one model. We ran different measures to see if the predictors of each galaxy had significant predictors by

first fitting linear models, and then secondly trying a random forest both yielded no significant results. Therefore, through this we found it very acceptable to say that these risks will be able to apply to the whole model instead of trying to make one for each individual solar system, since they all yielded the same results.

Additionally, we have tailored the product to consider various characteristics regarding the equipment that is used, as well as the behavior of the various solar systems that CQMC operates within. Some examples of this include the age of equipment, the quality of its installation, natural degradation of the equipment, poor maintenance, and the various risks that are associated within each system that CQMC operates within. These scenarios would be excluded from coverage in our pricing model.

We suggest to not include an insurance plan for Workers' Compensation. The data revealed an unrecognizable pattern that – with our limited resources – we were unable to distinguish. It is with our sincerest hope that CQMC will be more inclined to achieve high safety compliance and fairly compensate their workers in response.

1.3 ESG

Environmental considerations that must be accounted for in the case of CQMC is the company's history of environmental disputes within the Interstellar Court of Environmental Justice. With two lawsuits relating to environmental issues within the last 10 years totaling in damages of about \$ 115 million, environmental disputes may continue to be a problem for the company. Due to this, our product has made an exclusion in the case of environmental issues for CQMC, due to their less than ideal track record in the area.

1.4 Integration into ERM Framework

We are recommending the Cosmic Quarry Mining Corporation use our one fit model, as it seems as though each risk across solar systems are not significant predictors. Therefore, we can only look at the losses of the claims in full entirety. With this the model already considers all risk and hazard areas for business interruptions, equipment failure, cargo loss across the entire galaxy. We saw it fit to have different levels of deductibles to ensure that everyone can pay for their insurance instead of self-funding every claim, as this can cost much more than paying an insurance premium every year. Risks can avoid happening by paying their premiums and make sure that they are taken care of before they even submit a claim. This way they can be proactive about their risk. Also, workers should be conscience of the amount of exposure in their equipment, life of both their crew and of their equipment, as both can cause unforeseen risks from occurring. With this we want to

employ a safety compliance and ensure that all items, people, and equipment are up to space code this way we can minimize the number of risks, not only for our equipment, and operations, but also the workers that risk their lives for our mining endeavors.

Product Design

The proposed product will have a deductible policy limit that can vary with the preferences of CQMC. A table of policy limits or Amount of Insurance (AOI) is given below for each of the four hazard areas, with corresponding premiums. The deductibles for each of the four hazard areas are given below.

Hazard Area	Deductible (50 th percentile)
Equipment Failure	80000
Cargo Loss	2460000
Business interruption	3100000

Equipment Failure

AOI (Policy Limit)	Premium for individual policy
131704	3000
187719	3100
269208	3300

Cargo Loss

AOI (Policy Limit)	Premium for Individual Policy
40600000	708000
153498000	1662000
421499000	2831000

Business Interruption

AOI (Policy Limit)	Premium for Individual Policy
8915000	182000
14790000	229000
21529000	245000

2.1 Benefit Structures

Equipment Failure

We will lower the premium by a percentage (not yet determined) based on specific variables associated with claim counts and severity. For example, if maintenance intervals are less than or equal to the maintenance intervals specified by the manufacturer of the piece of equipment, then the discount might be applied.

Cargo Loss

The policy will be discounted by a percentage (not yet determined) for certain types of cargo vessels; specifically, cargo vessels with lower maximum volumes might have smaller premiums compared to vessels with higher maximum volumes.

Business Interruption

The policy will be discounted by a percentage (not yet determined) for values of variables associated with a lower frequency or severity. For example, stations with higher safety compliance scores might be given discounted premiums.

2.2 Coverage triggers

This insurance covers all losses within the three hazard areas of equipment failure, cargo loss, and business interruption, not including the exclusions listed below. The insurance will cover losses for one year specified in the contract. Losses outside of this year will not be covered. The insurance will only cover losses that provide sufficient evidence of loss

2.3 Exclusions

For all losses, sufficient evidence of losses must be provided. Speculative losses will not be covered.

Equipment failure that is due to the following will not be covered:

1. Age
 - a. Equipment 10 years or older years.
2. Poor installation
 - a. Based on manufacturer specified installation. Installation should be documented.
3. Natural degradation
4. Lack of maintenance
 - a. Maintenance should be documented by a third party.
5. Common events of the specified solar system

- a. Helionis Cluster:
 - i. Frequent collisions in asteroid cluster.
 - ii. Bad communication due to asteroid cluster
 - iii. Damage of relay satellites (or due to damage of these satellites) from cluster level fragmentation events.
- b. Bayesia System
 - i. Spikes in electromagnetic particle radiation
 - ii. Temperature extremes
 - iii. Elevated radiation levels.
- c. Oryn Delta
 - i. Low visibility
 - ii. Solar flares
 - iii. Damage due to inconsistent long-range communication
- d. Zeta:
 - i. Solar flare activity

Cargo Loss will not be covered if due to the following:

1. If cargo vessel is a quantum crate module or a long-haul value canister.
2. If loss is due to common environmental hazards (listed above for equipment failure).
3. If cargo is traveling a route with risk 5.
4. Due to negligence of operation.

Business Interruption will not be covered due to the following:

1. Government intervention and fines
2. Repercussions due to environmental damages
3. Lack of market demand and contracts
4. Voluntary interruptions
5. Extra and non-standard expenses incurred during interruption will not be covered (such as expenses incurred during resuming operation).

Summary of Pricing and Capital Modeling

To create a Pricing and Capital Model, we used the Central Limit Theorem to assume a normal distribution for the aggregate loss distributions for each hazard coverage. We computed the mean of each model by calculating the total expected loss per unit of exposure. Our exact process is listed in Appendix as Equation 1. In the table below are the means of the aggregate loss distributions for each hazard coverage.

Equipment Failure

(Deductible, Policy Limit)	Expected Aggregate Loss
(75523, 131704)	13708236
(75523, 187719)	13862476
(75523, 269208)	14692209

Cargo Loss

(Deductible, Policy Limit)	Expected Aggregate Loss
(2461368, 40583275)	2063871696
(2461368, 153497778)	4846633857
(3082606, 421498776)	8252980295

Business Interruption

We have no expected aggregate loss available as there is not a reliable number of policies/units that could result in a loss.

Risk Assessment

There are challenges related to each solar system that affect the risk that is encountered during the process of mining. There are overlaps in types of risk between the different star systems, but some have proven to be riskier than others due to the combination of different risk factors, while taking into consideration the low reward that is present from expanding operations in said systems.

From our research, we determined that the main risk related to the Helionis Cluster was the presence of asteroid clusters within the system. With these asteroid clusters, there are frequent collisions of debris, which negatively affect communications with the operations within that system. This is the only main issue that plagues the Helionis Cluster, but it is already in the process of being worked around with the presence of relay satellites. While this is something to keep in mind, it is not an issue that would profoundly affect the number of claims in the system, as it doesn't directly affect workers' performance.

When looking at the Bayesia System, the main risk that has the potential to significantly affect operations is the behavior of the system to experience temporary spikes of electromagnetic radiation. This is a challenge of a different scale compared to the Helionis Cluster, as this poses significant risk to the workers present in the system. Spikes

in electromagnetic radiation can be associated with a higher number of workers' compensation claims, which would be more costly to ensure operations were expanded into the system. This problem extends into the first planet of the Bayesia System, the only one suitable for mining operations. This planet contains elevated ambient radiation levels combined with frequent temperature extremes. With such little reward associated with all these different challenges, the risk doesn't seem to be worth taking on for this system.

The risks associated with the Oryn Delta are very much related to its red dwarf star. With this red dwarf star, there is low luminosity in the system, along with unpredictable solar flares. Pairing this with unreliable long-distance communications makes this system vulnerable to high levels of risk, but it does have much higher upside compared to the other systems. Due to the dense concentration of metals in the asteroid ring of the delta, it seems a worthy tradeoff to make due to the importance of these metals for advanced manufacturing.

For expansion of future operations, the prospective destinations of the Zeta and the Epsilon solar systems have risks that provide a significant barrier of entry for mining operations. The Zeta solar system contains solar flare activity that operates at a constant moderate rate, providing a challenge when it comes to the protection of mining equipment. Equipment claims would be significantly affected by this characteristic of the solar system. Additionally, the outer belt of the system contains various astronomical bodies that provide a challenge to ships navigating through the system to deliver supplies and to provide maintenance to equipment. This characteristic of the Zeta system would have a direct effect on cargo claims, making it important to take into consideration. The Epsilon system is not free of obstacles either. The system contains planets with a dense and corrosive atmosphere, and its planets have extreme weather fluctuations. This makes having operations on these planets hard to maintain, and it can affect claims on equipment when it comes to maintenance. Another characteristic of the Epsilon system is the high levels of UV radiation that affect the planets. This requires significant investment in equipment that can handle the behavior of these planets. Overall, the resources and rare metals that are present in the planets of these two systems provide massive manufacturing potential, but the presence of these risks that are present in these systems are something to consider when the topic of expansion of operations comes up.

Assumptions

We will now discuss the key assumptions that our team made in creating this project. Most generally, we assumed that the year of data given to us was complete and not an outlier compared to past years or future years. This allows us to make reasonable

estimates of future years and so necessary to make our product. Furthermore, it lets us say the number of policies stays relatively constant as well as the likelihood of claims.

Additionally, when cleaning each claims dataset, the categorical variables given to us had many unclean values that included random symbols following an underscore. We removed this part of the values assuming they were mistakenly added on to the policy_id, cargo_type, etc. For any numerical variables, we assumed values that were outside of the specified set of possible values to be incorrect data and designated them as NA without removing each entry.

Some other assumptions we made about the Galaxy General Insurance Company include that our overall expenses are comparable to insurance companies on Earth in the 21st century. Without knowledge of our company's overall size, we thought it was prudent to assume similarity to existing companies that we know. We also set our target profit at 10% of our revenue and the amount of capital we have on hand to be approximately 250% of our total claim amount from our given year of data. Our target profit is reasonable for large insurance companies like ours and our capital amount is conservative estimate that allows us to be able to comfortably cover our losses even in more extreme years.

Data and Data Limitations

For this project, the main data sources we used in addition to the provided data were the Loss Data Analytics Textbook and the Basic Ratemaking Textbook.

Through several linear regression models between proposed risk factors and claim amount and claim count, we found no significant predictors. We believe, likely because of the many variables interacting and similarity of observed values, that no simple predictable relationship exists between our explanatory and response variables. We validated this further through a random forest analysis, which assigned very similarly low significance to each of our response variables.

The claims datasets provided required extensive cleaning, causing our number of usable entries for certain variables to decrease greatly. As a result, our predictions and statistics are less likely to estimate our parameters well. We viewed this as a necessary evil, as retaining these likely erroneous values would throw off our predictions more than their omission.

The workers' compensation dataset is very case-to-case dependent, and we have a lack of information on intergalactic workers' compensation regulations and policies. As such, we used the mean or expected value as the primary estimator for loss amount in our aggregate loss distribution.

In terms of the usage of artificial intelligence for this project, our team only used AI to aid in understanding complex actuarial topics and concepts that we were not familiar with. AI was not used to generate any parts of our report, including our writing, visuals, etc.

Sources

- Kanaris, Peter E. “ANALYTICAL APPROACH TO BUSINESS INTERRUPTION, EXTRA EXPENSE, AND CIVIL AUTHORITY COVERAGE ISSUES.” *Tort Trial & Insurance Practice Law Journal*, vol. 43, no. 1, 2007, pp. 113–38. *JSTOR*, <http://www.jstor.org/stable/25763877>. Accessed 13 Mar. 2026.
- “Loss Data Analytics.” *Github.io*, 2022, openacttexts.github.io/Loss-Data-Analytics/.
- Gannon, Fcas, et al. *BASIC RATEMAKING*. 2016.

Appendix

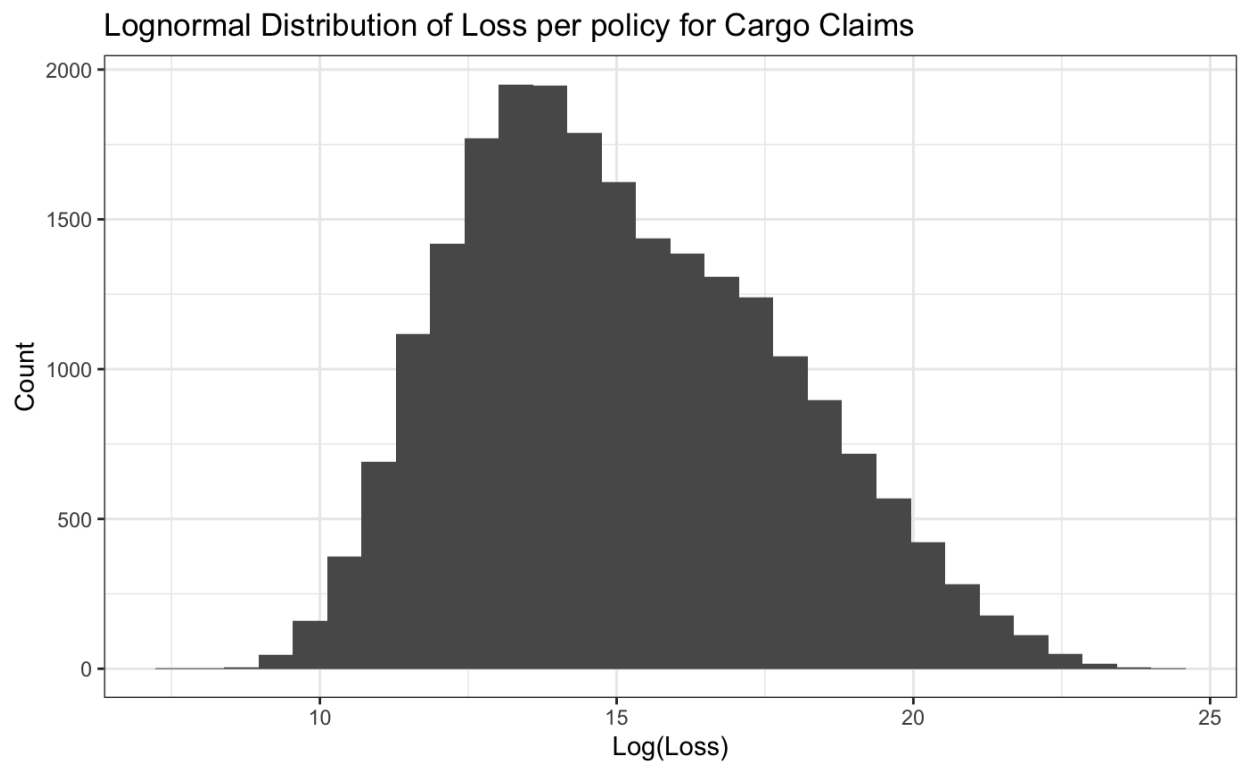


Figure 1: Lognormal Distribution of Loss Per Policy for Cargo Claims

Code:

```
cargo_claims_freq <- cargo_claims_freq %>%
```

```
  filter(!is.na(exposure))
```

```
sum(cargo_claims_freq$exposure)
```

```
new_cargo_claims <- cargo_claims_sev %>%
```

```
  group_by(policy_id) %>%
```

```
  summarize(total_claim_amount = sum(claim_amount)) %>%
```

```
  right_join(cargo_claims_freq, by = join_by(policy_id)) %>%
```

```
  filter(claim_count != 0) %>%
```

```
  mutate(total_claim_amount = ifelse(is.na(total_claim_amount), 0, total_claim_amount),
```

```

    loss = (claim_count / exposure) * total_claim_amount) %>%
filter(!is.na(loss))

new_cargo_claims %>%

ggplot(aes(x = log(loss))) +
geom_histogram()+
labs(x = "Log(Loss)",
     y = "Count",
     title = "Lognormal Distribution of Loss per policy for Cargo Claims")+
theme_bw()

```

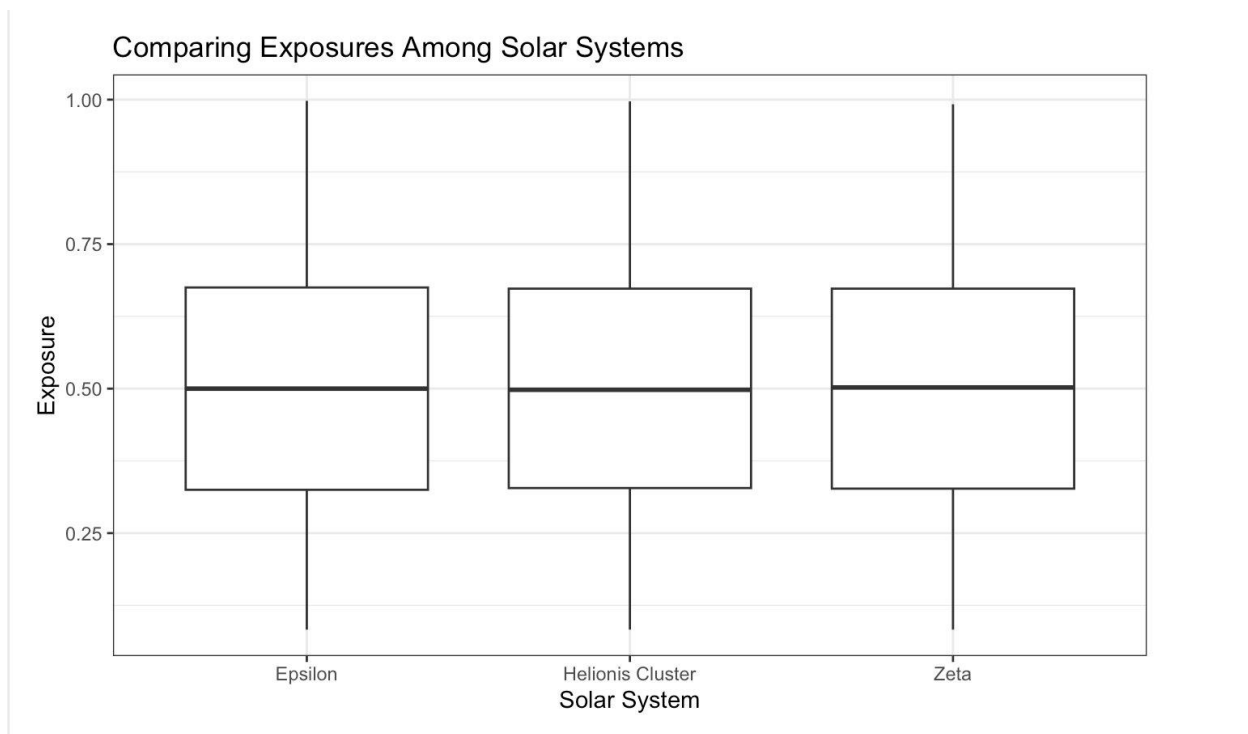


Figure 2: Comparing Exposures Among Solar Systems

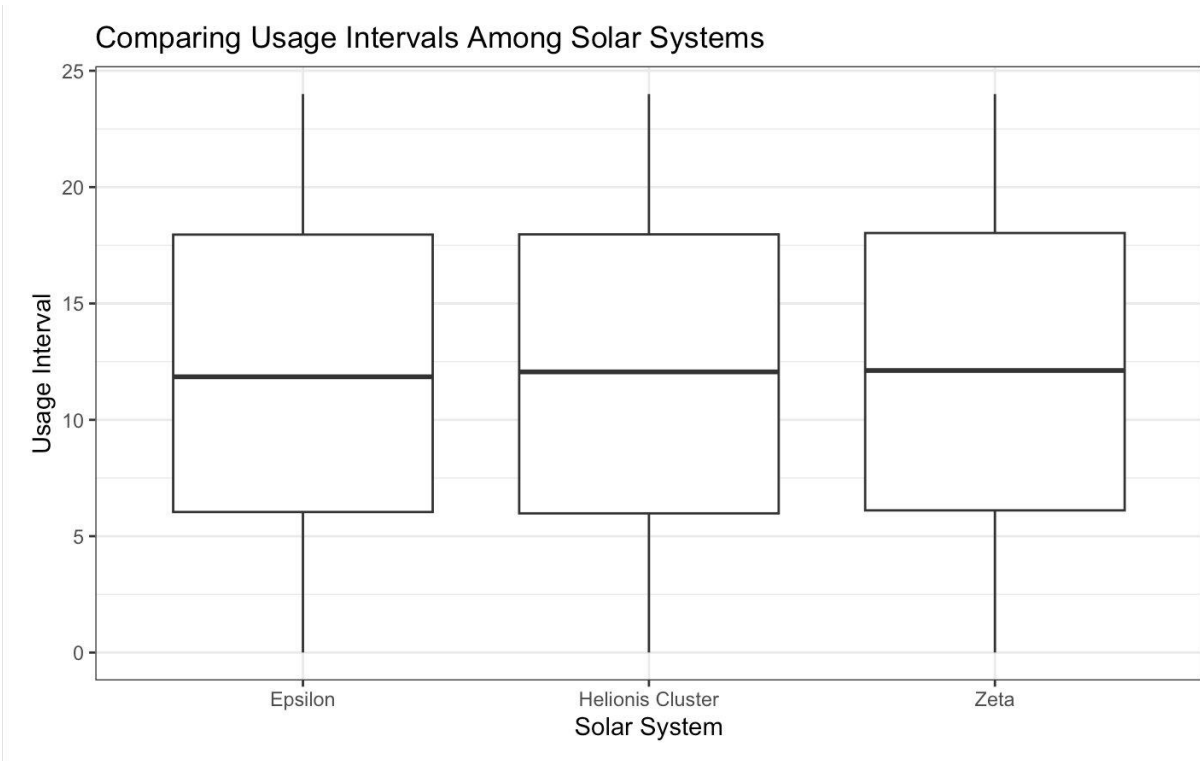


Figure 3: Comparing Usage Across Solar Systems

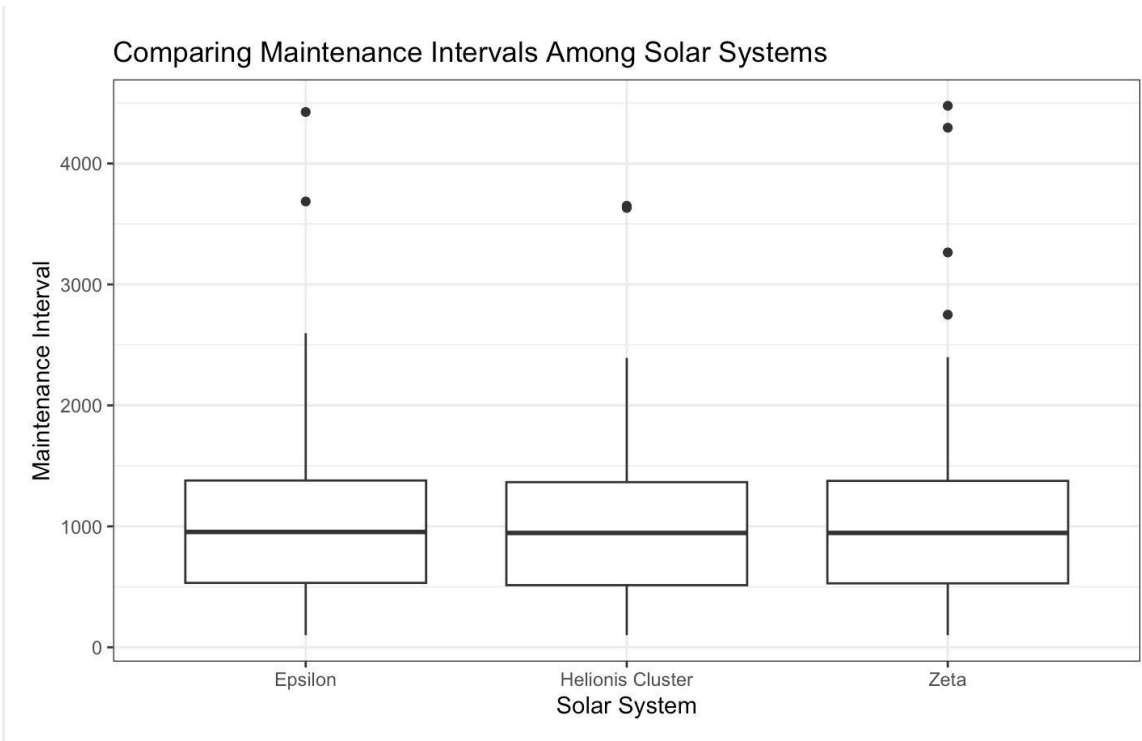


Figure 4: Comparing Maintenance Intervals Among Solar Systems

Table 1: Loss Distribution with Normal Overlaid on the Equipment Failures Claims

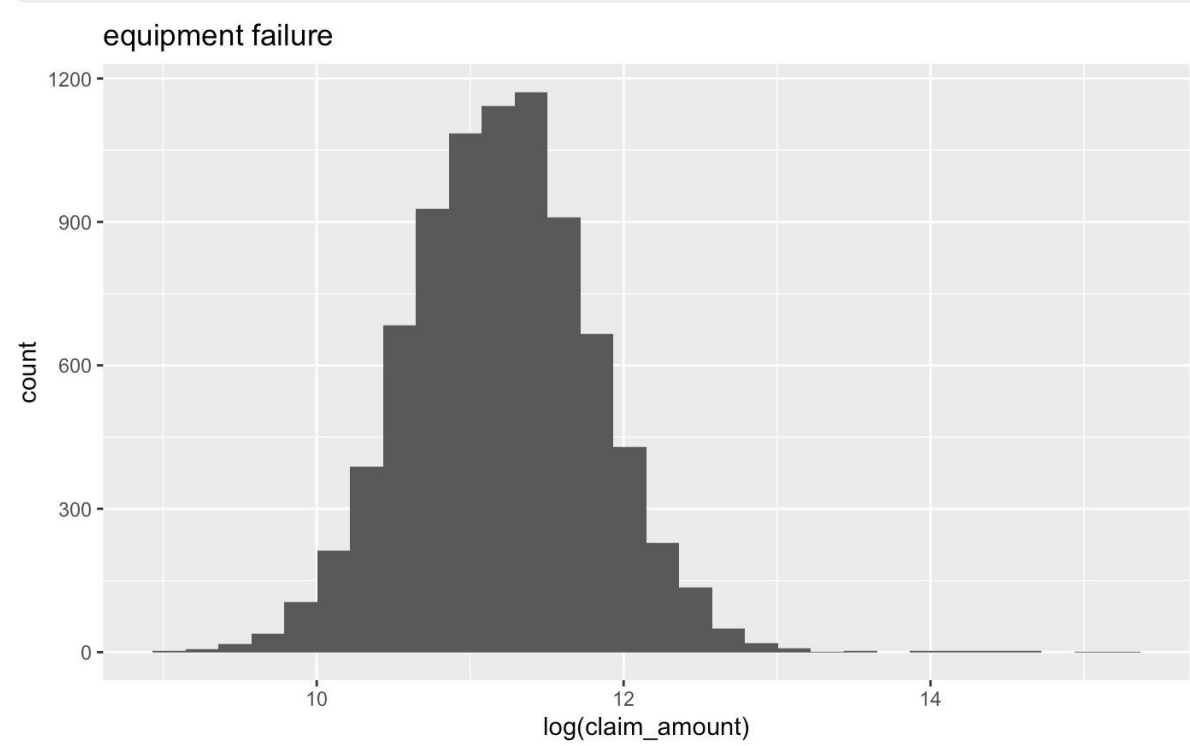
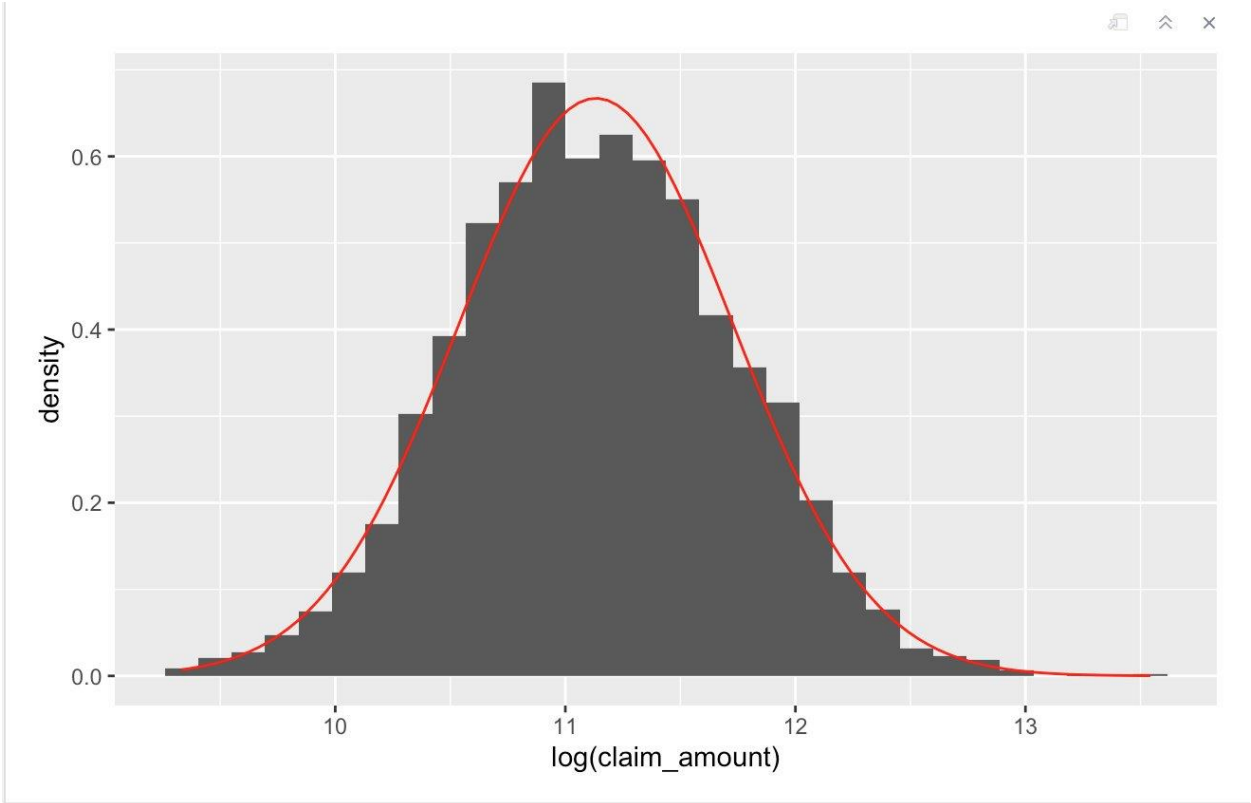


Figure 5: Equipment Failure Loss Distribution

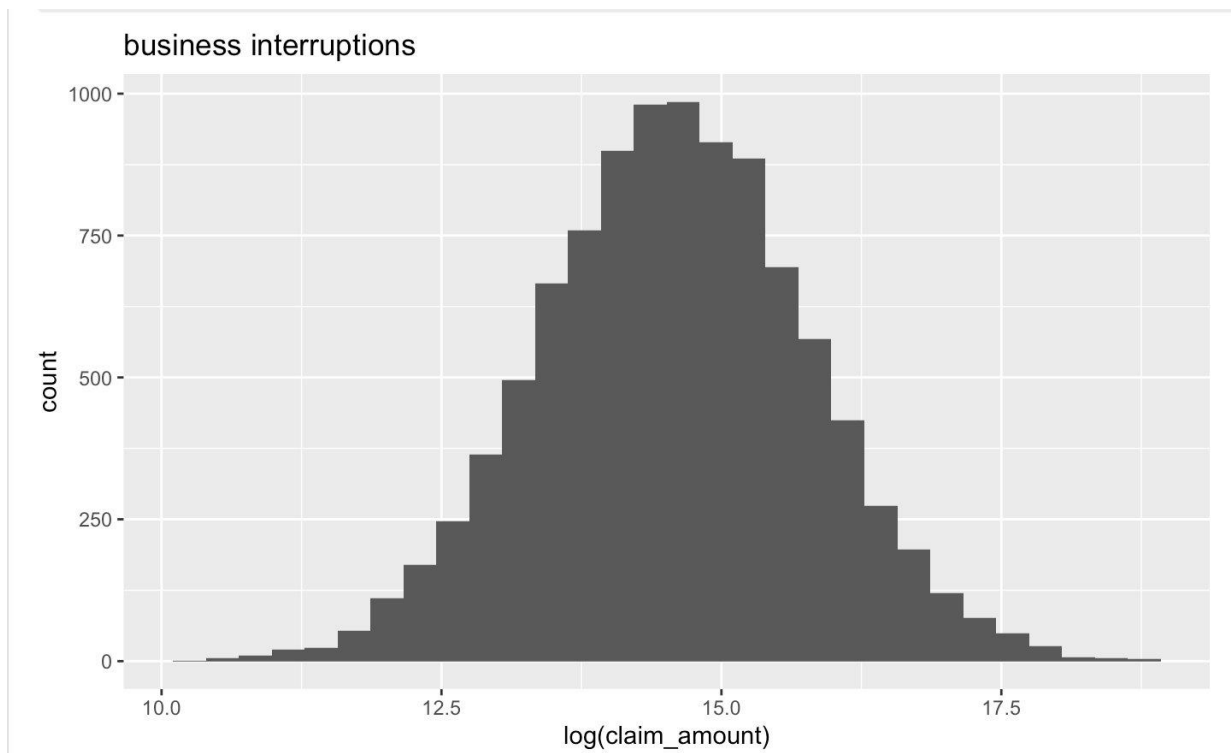


Figure 6: Business Interruptions Loss Distribution

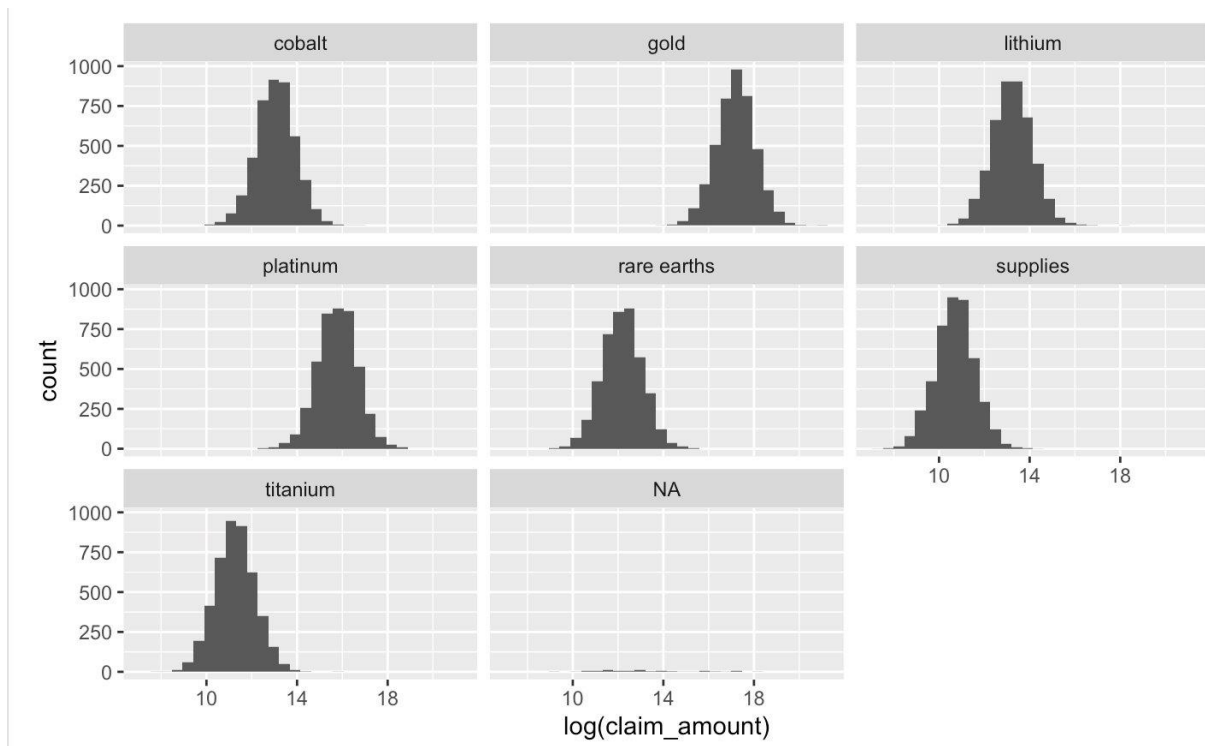


Figure 7: Metals influencing Claim Amounts

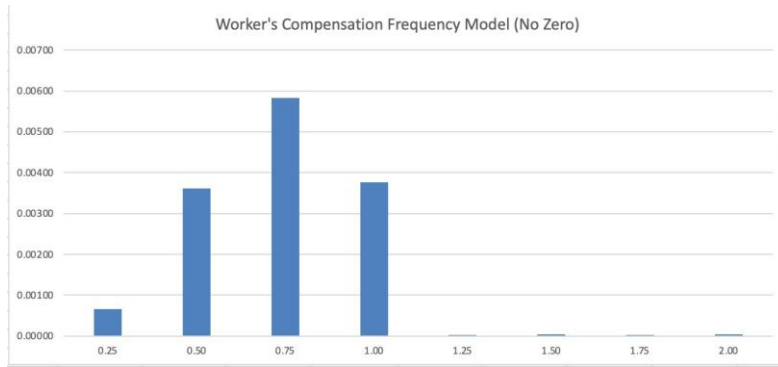


Figure 8: Workers Compensation Frequency Model (No Zero)

Table 2: Workers Comp Frequency Table

Interval	Count	Proportion
0	88058	0.926770228
0.25	387	0.004072998
0.5	1754	0.018460049
0.75	2866	0.030163341
1	1643	0.017291825
1.25	86	0.000905111
1.5	78	0.000820914
1.75	85	0.000894586
2	43	0.000452555
2.25	4	4.20982E-05
2.5	4	4.20982E-05
2.75	5	5.26227E-05
3	2	2.10491E-05
3.25	1	1.05245E-05
Total	95016	1

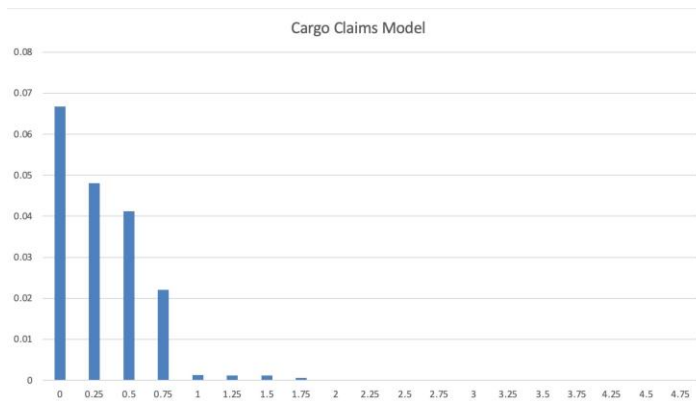


Figure 9: Cargo Claims Model

Table 3: Cargo Claims

Interval	Count	Proportion	Total
0	8282	0.066719836	124131
0.25	5958	0.04799768	
0.5	5112	0.041182299	
0.75	2737	0.022049287	
1	160	0.001288961	
1.25	150	0.001208401	
1.5	150	0.001208401	
1.75	68	0.000547808	
2	5	4.028E-05	
2.25	6	4.8336E-05	
2.5	8	6.4448E-05	
2.75	5	4.028E-05	
3	1	8.05601E-06	
3.25	3	2.4168E-05	
3.5	5	4.028E-05	
3.75	2	1.6112E-05	
4.25	3	2.4168E-05	
4.5	3	2.4168E-05	
4.75	2	1.6112E-05	

How we came up with Pricing Model, and Product Design:

Product Design

- X_i is the random variable representing the individual claim amount.
- Y_i is the loss of an individual claim.
- d is the deductible.
- L is the policy limit.

$$Y_i = \begin{cases} 0, & \text{if } X < d \\ X - d, & \text{if } d \leq X \leq L + d \\ L, & \text{if } X > L + d \end{cases}$$

Calculating expected value of loss.

$$\begin{aligned} E[Y_i] &= 0 \cdot \Pr(X < d) + E[X - d \mid d \leq X \leq L + d] \cdot \Pr(d \leq X \leq L + d) + E[L \mid X > L + d] \cdot \Pr(X > L + d) \\ &= E[X \mid d \leq X \leq L + d] \cdot \Pr(d \leq X \leq L + d) - d \cdot \Pr(d \leq X \leq L + d) + L \cdot \Pr(X > L + d) \end{aligned}$$

Finding conditional expected value of the claim amount.

- $E[X_i \mid d \leq X_i \leq L + d]$ found through filtering data to only include losses above d and below $L + d$ and then taking the average.
- d is the 50th percentile.
- L is offered as 80th, 90th, or 95th percentile depending on each solar system's unique circumstances.

Accounting for probability a policy has at least one claim. Then calculating total premium.

- N is the total number of policies with at least one claim.
- p is the proportion of policies with at least one claim.
- $E[Z_i] = p \cdot E[Y_i]$

$$\text{Total loss} = E[Z_T] = N \cdot E[Z_i]$$

$$\text{Profit} = 0.05E[Z_i]$$

$$\text{Single Policy Premium} = \text{Profit} + \text{Loss on a Single Policy}$$

$$\begin{aligned} \text{Total Premium} &= N \cdot (\text{Single Policy Premium}) \\ &= 0.05 \cdot N \cdot E[Z_i] + N \cdot E[Z_i] \\ &= 0.05E[Z_T] + E[Z_T] \end{aligned}$$

Equation 2: Random Forest on Business Interruptions Severity Data

parsnip model object

Ranger result

Call:

```
ranger::ranger(x = maybe_data_frame(x), y = y, num.trees = ~500,  
num.threads = 1, verbose = FALSE, seed = sample.int(10^5, 1))
```

Type:	Regression
Number of trees:	500
Sample size:	2443
Number of independent variables:	10
Mtry:	3
Target node size:	5
Variable importance mode:	none
Splitrule:	variance
OOB prediction error (MSE):	2280564484
R squared (OOB):	0.198649

Equation 3: Random Forest on Equipment Failure Severity Data

parsnip model object

Ranger result

Call:

```
ranger::ranger(x = maybe_data_frame(x), y = y, num.trees = ~500,  
num.threads = 1, verbose = FALSE, seed = sample.int(10^5, 1))
```

Type:	Regression
Number of trees:	500
Sample size:	36993
Number of independent variables:	8
Mtry:	2
Target node size:	5
Variable importance mode:	none
Splitrule:	variance
OOB prediction error (MSE):	0.0493922
R squared (OOB):	-0.005488881

Equation 4:Random Forest on Equipment Failure Frequency Data

```
parsnip model object

Ranger result

Call:
  ranger::ranger(x = maybe_data_frame(x), y = y, num.trees = ~500,
num.threads = 1, verbose = FALSE, seed = sample.int(10^5,      1))

Type:                                Regression
Number of trees:                      500
Sample size:                          2443
Number of independent variables:      10
Mtry:                                  3
Target node size:                     5
Variable importance mode:             none
Splitrule:                            variance
OOB prediction error (MSE):           2280564484
R squared (OOB):                      0.198649
```

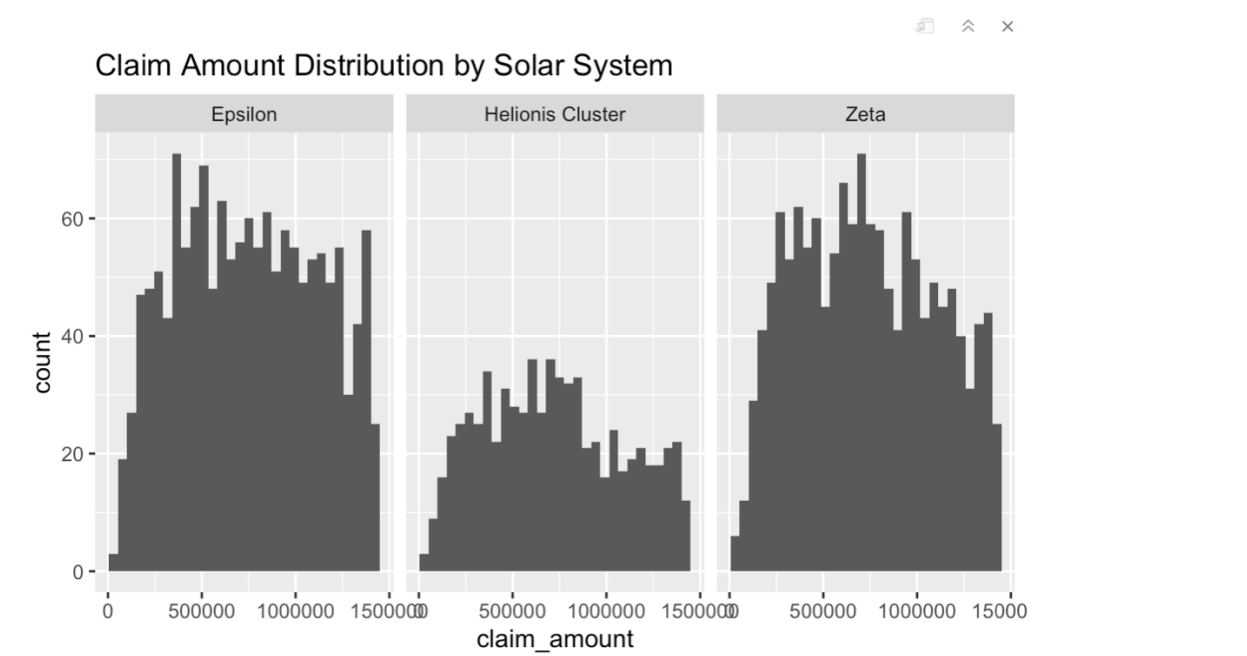


Figure 10:Claim Amount Distribution by Solar System